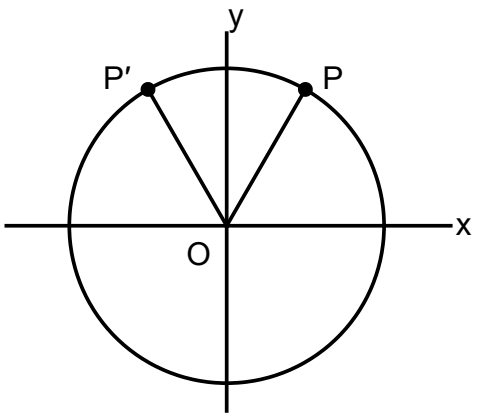
MATHS AA HL · Trigonometric symmetry and identities

Question booklet · 25 minutes · 18 marks

Answer all questions. Work without a calculator. Show your reasoning. Give exact answers or round to three significant figures unless directed otherwise. Write on separate paper.

1. Reflections and a hidden root [10]

For $0 \le x \le 2\pi$ define $F(x) = \sin x + \sin(\pi - x) + \cos x - \cos(\pi - x)$. Work without a calculator. The diagram illustrates the reflection for a point P in the first quadrant.



P′ is the reflection of P in the vertical axis.

(a) Use unit-circle symmetry to simplify $F(x)$. **[3]**

(b) Solve $F(x) = 2$ exactly, retaining both endpoints where appropriate. **[3]**

(c) Find all real k such that $F(x) = k$ has exactly three distinct solutions on the stated closed interval. Justify completeness. **[4]**

2. Which transformations preserve a trigonometric equation? [8]

Let $g(x) = \sin^2 x - \cos^2 x$ for real x. Work without a calculator.

(a) Show that $g(\pi - x) = g(x)$ and $g(x + \pi/2) = -g(x)$. **[4]**

(b) Solve $g(x) = 1/2$ on $0 \le x < 2\pi$. A student argues that shifting each solution by $\pi/2$ supplies four more. Evaluate the argument. **[4]**

End of question booklet · 18 marks